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Quantum Random Access

A short-oracle separation and QRAM-free TSP

Abstract. We prove two statements about random access and fixed quantum wiring. First, relative to independent random oracles with $O(\log N)$-bit inputs and outputs, a Boolean function has an exact $O(N \log N)$-gate random-access algorithm, while every ordinary quantum oracle circuit of size at most $cN^2/\log N$ has average success at most $1/2 + N^{-10}$. The lower bound allows coherent oracle queries and unlimited workspace. It is distributional over the input and oracles, against an oracle-independent circuit description. Its main ingredient is a short-block decoding argument applied to a modified Chen–Wang construction.

Second, a fixed set family of size M and full-chain density p gives a QRAM-free TSP algorithm with $O^*(M/\sqrt{p})$ gates and $O^*(M)$ qubits. The dynamic program has fixed dependencies; only its polynomial-size input is relabeled coherently. An existing extremal-family theorem yields $O^*(1.784^n)$ gates and $O^*(1.430^n)$ qubits. All gates in this construction are exact over a fixed finite gate set. The two results are compatible: a worst-case addressing obstruction does not make QRAM necessary for every structured algorithm.

Research status. This remains an internally audited research draft, not an independently refereed result. The substantive external inputs are the Ben-Aroya–Regev–de Wolf random-access-code bound and the Andoni–Dallant–Kozma–Yu extremal-family theorem. No priority claim is made for the QRAM-free reformulation.

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1 Results and computational models

An *ordinary quantum circuit* has classical input on designated basis-state wires, zero ancillas, a fixed list of bounded-arity gates, and computational-basis measurements of designated output qubits. Gate locations are prescribed; a register cannot select a wire for free. Size is total gate count, with no geometric restriction on connections. Fixed-location classical controls are allowed and may be deferred coherently. An uncharged controller that selects arbitrary memory locations is not allowed.

In Part I, both models also have unit-cost XOR-query gates

$$U_f|u, z\rangle = |u, z \oplus f(u)\rangle.$$

Every query and response has $O(\log N)$ bits, and its entire register counts among the qubits touched by that gate. Queries may be coherent. The circuit description is independent of the sampled oracle tables; all input- or oracle-dependent preprocessing is charged. Workspace is unrestricted. The lower bound also allows any fixed auxiliary state independent of the input and oracles.

Only the stronger model has the unit-cost random-access gate (RAG)

$$\mathsf{Read}|i, z, x\rangle = |i, z \oplus x_i, x\rangle. \tag{1}$$

This is an addressed read of the explicit input array. An addressed swap can implement it with a zero buffer, a CNOT, and a swap back.

Theorem A (Short-random-oracle separation)**.** *For every sufficiently large power of two N, there is an explicitly defined Boolean function $F_N^{G,H} : \{0,1\}^N \to \{0,1\}$, relative to independent uniform random functions G, H with $O(\log N)$-bit inputs and outputs, with the following properties.*

A uniform RAG oracle circuit computes $F_N^{G,H}$ exactly for every input and every oracle pair, using $O(N \log N)$ gates and $O(N \log^2 N)$ qubits. In contrast, for an absolute $c > 0$, every ordinary quantum oracle circuit C of at most $cN^2/\log N$ gates, chosen independently of G, H, satisfies

$$\mathbb{P}_{G,H,X,\mathrm{meas}}[C^{G,H}(X) = F_N^{G,H}(X)] \le \tfrac{1}{2} + N^{-10}, \tag{2}$$

where X is uniform and independent of the oracles.

For Part II, n is the number of cities, and $O^*(\cdot)$ suppresses only factors polynomial in n and in the bit length b of the weights. All logarithms are base two.

Theorem B (QRAM-free TSP)**.** *For an explicit n-vertex directed graph with integer edge weights of polynomial bit length, there is a uniform bounded-error quantum algorithm using*

$$O^*(1.784^n) \text{ ordinary gates} \qquad \text{and} \qquad O^*(1.430^n) \text{ ordinary qubits}.$$

It outputs an optimum Hamiltonian tour and its value, or reports that no tour exists. It uses neither RAGs nor a free input-query oracle. Its gates can be chosen from a fixed finite bounded-arity gate set and implemented exactly.

Scope. Theorem A is a quantum-circuit analogue of the short-oracle RAM/Turing-machine separation of Chen–Wang [1], using a different hard function. Its upper-bound algorithm is classical on basis-state inputs, so it does not separate QRAM from a model already granted free classical input addressing. Nor does it establish an almost-every-oracle bound against oracle-dependent nonuniform circuits. Theorem B is unrelativized but retains exponential *coherent* workspace. Neither result is a QRAM lower bound for TSP.

PART I

A short-oracle separation

2 The hard function

Put $N = 2^m$ and $r = \lambda = 100m$. Take independent uniform random functions

$$G : \{0, \dots, N-1\} \times \{0,1\}^\lambda \times [r] \longrightarrow [N],$$
$$H : \{0, \dots, N-1\} \times \{0,1\}^\lambda \times \{0,1\}^r \longrightarrow \{0,1\}^\lambda.$$

Here $[d] = \{1, \dots, d\}$; the output of G has an m-bit encoding. Unused encodings of $[r]$ receive independent random outputs as well. Starting from $s_0 = 0^\lambda$, define, for $0 \le t < N$,

$$\begin{aligned} i_{t,j} &= G(t, s_t, j) \qquad (j \in [r]), \\ v_t &= (x_{i_{t,1}}, \dots, x_{i_{t,r}}), \\ s_{t+1} &= H(t, s_t, v_t). \end{aligned} \tag{3}$$

The output $F_N^{G,H}(x)$ is the first bit of s_N. Every oracle layer is accessible from the beginning; t is an input field, not an access restriction.

The longest oracle gate touches

$$w = m + \lambda + r + \lambda = 301 \log N \tag{4}$$

qubits. An H-query address has length $k = m + \lambda + r = 201 \log N$.

Proposition 2.1 (Random-access upper bound)**.** *The recurrence* (3) *has a uniform exact RAG implementation with $O(N \log N)$ gates and $O(N \log^2 N)$ qubits.*

Proof. At stage t, make r queries to G into fresh index registers, use r RAGs to obtain v_t, and query H into a fresh register for s_{t+1}. Keep all intermediate registers at fixed locations. The seed s_t is an unchanged input to the queries and need not be copied for each one.

There are Nr calls to each of G and Read, and N calls to H. Preparing the stage and position constants once takes $O(Nm + r \log r)$ ordinary gates. The index registers occupy $Nrm = O(N \log^2 N)$ qubits; everything else uses no more space. To clean the workspace, copy the final bit and reverse the computation, at constant-factor overhead. □

Lower-bound strategy. A block of $\Theta(N/\log N)$ ordinary gates can touch only a small fraction of the input-sized memory. A quantum random-access-code bound prevents that block from recovering a fresh batch of $\Theta(\log N)$ input bits. We couple the random oracles to a planted chain and delay one transition per block in a comparison execution. This makes the required batch fresh at every fixed block boundary. A coherent-query hybrid then compares that execution with the real one. No query transcript is measured or assumed to be classical.

3 The lower bound

Lemma 3.1 (Batch retrieval)**.** *Let X be uniform on $\{0,1\}^N$, and encode it into at most $N/8$ qubits. If $J = (J_1, \dots, J_r)$ is an independent uniform element of $[N]^r$, with $r = 100 \log N$, then any decoder has probability at most*

$$\varepsilon_N = C(4/5)^r \tag{5}$$

of outputting the entire tuple $X_J = (X_{J_1}, \dots, X_{J_r})$, for an absolute C and sufficiently large N.

The encoding may be mixed and inefficient. The decoder may also know arbitrary classical side information Z independent of X; the encoding may depend on (X, Z), but J must remain independent of $(X, Z,$ encoding$)$.

Proof. Ben-Aroya–Regev–de Wolf [2, Definition 1 and Theorem 2] show that, from at most $N/8$ qubits, recovering a uniform d-subset succeeds with average probability at most $C(3/4)^d$ when N/d is sufficiently large. This is their bound with parameter $\eta = 2$, and it allows mixed encodings.

Let D be the number of distinct coordinates in J. Conditional on $D = d$, these form a uniform d-subset. The multiplicities and order can be sampled by the decoder independently of X, given that subset, so success is at most $C\mathbb{E}[(3/4)^D]$. If D_j counts distinct coordinates in the first j draws, then

$$\mathbb{E}[(3/4)^{D_j} \mid D_{j-1}] = (3/4)^{D_{j-1}} \left(\tfrac{3}{4} + \frac{D_{j-1}}{4N} \right).$$

Iterating and using $D_{j-1} \le r$ gives

$$\mathbb{E}[(3/4)^D] \le \left(\tfrac{3}{4} + \frac{r}{4N} \right)^r \le (4/5)^r.$$

The external theorem applies uniformly because $N/d \ge N/r \to \infty$. With side information, condition on $Z = z$; X is still uniform. Apply the same bound to the conditional encoding and average over z. □

Lemma 3.2 (Coherent-query hybrid)**.** *Consider real and comparison executions of the same circuit. At oracle call j, let B_j contain the inputs where the two oracle tables differ, and let $\mu_j(B_j)$ be the query probability on that set in the comparison execution. The comparison oracle may vary with the gate position. Then*

$$\mathbb{E}\ \mathrm{D}_{\mathrm{tr}}(\rho_{\mathrm{real}}, \rho_{\mathrm{comp}}) \le 2 \sum_j \sqrt{\mathbb{E}\, \mu_j(B_j)}, \tag{6}$$

where $\mathrm{D}_{\mathrm{tr}}(\rho, \sigma) = \frac{1}{2}\|\rho - \sigma\|_1$. The expectation may be over any jointly sampled classical variables specifying the executions.

Proof. Fix all sampled variables and purify the initial state. If XOR-query unitaries U, V differ only on B, and P_B projects onto those query inputs, then

$$\|(U - V)|\psi\rangle\| = \|(U - V)P_B|\psi\rangle\| \le 2\sqrt{\mu(B)}.$$

Telescope the full gate products with a comparison prefix and a real suffix:

$$U_S \cdots U_1 - V_S \cdots V_1 = \sum_{j=1}^{S} U_S \cdots U_{j+1}(U_j - V_j)V_{j-1} \cdots V_1.$$

Thus each discrepancy is evaluated on the comparison state. Sum the norm bounds; trace distance is no larger, even after discarding the purification. Finally average and use concavity of the square root. This is the usual oracle-hybrid estimate [3]. □

3.1 Planting the chain without changing the oracle distribution

Independently sample a uniform input X, uniform base oracles G°, H°, uniform seeds $s_1, \dots, s_N \in \{0,1\}^\lambda$, and uniform tuples $I_0, \dots, I_{N-1} \in [N]^r$. Set $s_0 = 0^\lambda$ and

$$a_t = (t, s_t, X_{I_t}).$$

From the base tables, form final tables G^*, H^* by the replacements

$$G^*(t, s_t, j) = (I_t)_j \quad (j \in [r]), \qquad H^*(a_t) = s_{t+1} \quad (0 \le t < N). \tag{7}$$

All other entries retain their base values. Different stages lie in disjoint layers.

We also need intermediate snapshots. Let $\mathcal{O}_{-1} = (G^\circ, H^\circ)$, and let $\mathcal{O}_t$ program the G-layers through stage t and the H-layers through stage $t-1$:

Snapshot	Programmed G stages	Programmed H stages
$\mathcal{O}_{-1}$	none	none
$\mathcal{O}_t$	$0, \dots, t$	$0, \dots, t-1$
$\mathcal{O}^* = (G^*, H^*)$	$0, \dots, N-1$	$0, \dots, N-1$

The current H-transition is still unprogrammed in $\mathcal{O}_t$.

Lemma 3.3 (Uniform snapshots and fresh variables)**.** *Each single snapshot $\mathcal{O}_t$, as well as $\mathcal{O}^*$, consists of independent uniform oracle tables and is independent of X. Moreover,*

$$F_N^{G^*,H^*}(X) = (s_N)_1. \tag{8}$$

For $t \ge 1$, the pair (s_t, I_t) is independent of X, the snapshots through $\mathcal{O}_{t-1}$, and a_{t-1}, jointly. For $t = 0$, I_0 is fresh and s_0 is fixed.

Proof. Fix $X = x$ and expose the final layers in the order $G_0, H_0, G_1, H_1, \dots$. Before exposing G_t, the earlier exposed layers determine s_t. Hence the positions (t, s_t, j) are fixed, and their replacement values are fresh independent uniform outputs. Replacing these entries of an independent uniform base layer leaves the entire layer uniform, independently of the exposed past.

Next, $a_t = (t, s_t, x_{I_t})$ is fixed by the exposed information. Replacing its entry in the new H_t layer by the fresh uniform s_{t+1} again leaves that layer uniform and independent of the past. Thus all final layers have the product-uniform distribution for every x, proving independence of X. For $\mathcal{O}_t$, stop programming after G_t; all remaining layers are untouched independent base layers. The recurrence (7) proves (8).

Finally, the snapshots through $\mathcal{O}_{t-1}$ and the address a_{t-1} are functions of the base tables, X, seeds through s_{t-1}, and tuples through I_{t-1}. They do not use the independently sampled (s_t, I_t). □

The conditioning rule. A *single* old snapshot is independent of X; its entire history need not be. Snapshot differences expose input-dependent addresses. The proof below gives the decoder just one old snapshot, charges the one required transition address to its encoding, and never gives away the history.

3.2 A short block cannot retrieve its current batch

Let C have at most NL gates, where

$$L = \left\lfloor \frac{N}{16w} \right\rfloor, \qquad w = 301 \log N. \tag{9}$$

Pad with identities and partition the circuit into N consecutive blocks of L gates. The real execution uses O^* throughout; the comparison execution uses O_t throughout block t. These time-dependent snapshots are only a proof device.

Let W_t be the fixed set of all qubits touched anywhere in block t. Then

$$|W_t| \le wL \le N/16. \tag{10}$$

There is no bound on the qubits stored elsewhere.

Lemma 3.4 (Current-transition query mass)**.** *Immediately before any particular H-query in comparison block t, the expected query probability on $a_t = (t, s_t, X_{I_t})$ is at most ε_N.*

Proof. Suppose first that $t \ge 1$. Let $Z = O_{t-1}$ be the entire old pair of oracle tables. Condition on $Z = h$; by Lemma 3.3, X remains uniform.

The encoding. Let ρ_t be the entering comparison state for a sampled planting outcome. Encode x by the joint classical–quantum state

$$\omega_x^h = \mathbb{E}\Big[|a_{t-1}\rangle\langle a_{t-1}| \otimes \mathrm{Tr}_{\overline{W_t}} \rho_t \,\Big|\, X = x,\ Z = h \Big]. \tag{11}$$

This averages over earlier planting variables while *preserving the correlation* between the address register and the reduced state. It uses at most

$$|W_t| + |a_{t-1}| \le N/16 + 201 \log N \le N/8$$

qubits, for sufficiently large N. No efficient preparation is required. The fresh pair (s_t, I_t) is independent of (X, Z, ω_X^Z): none of the snapshots already used, the entering state, or a_{t-1} depends on it.

The decoder. Give a decoder the encoding, the old table h, and a fresh uniform tuple $J \in [N]^r$. It reads the classical address register, samples an independent uniform $s \in \{0,1\}^\lambda$, and constructs

$$h' = h\big[H(a_{t-1}) \leftarrow s; \quad G(t, s, j) \leftarrow J_j \ (j \in [r])\big]. \tag{12}$$

This is exactly the update from O_{t-1} to O_t, with (s, J) in place of the identically distributed fresh (s_t, I_t).

Knowing the entire classical h', the unbounded decoder simulates the block up to the chosen query. Every gate acts on W_t, so its reduced state suffices; entanglement with untouched qubits is irrelevant to these local measurement probabilities. Conditional on each encoded address, the simulation uses precisely the corresponding oracle update, so the joint distribution is correct.

Measure the H-query input and output its last r bits. Whenever the query equals (t, s, X_J), this outputs the entire requested tuple X_J. Hence the decoder's success is at least the current-transition query probability. Lemma 3.1, conditioned on h and then averaged, gives the bound.

For $t = 0$, condition on the base table O_{-1} and encode only the reduced initial state on W_0. Use the fixed seed s_0 and update only $G(0, s_0, j) \leftarrow J_j$. The same argument applies. □

The old table is free side information because it is independent of X; the previous transition address is not free and is included in (11). The query measurement belongs only to the hypothetical decoder. Neither actual execution is measured along the way.

3.3 Future locations and the final hybrid

For either oracle, any future programmed location in stage $u > t$ has seed field s_u. These future seeds are independent of the comparison state in block t. If μ denotes that state's query distribution, then

$$\mathbb{E}\,\mu(\text{future programmed inputs}) \le \sum_{u>t} 2^{-\lambda}\mathbb{E}\,\mu(\text{stage} = u) \le 2^{-\lambda}. \tag{13}$$

Here we simply ignore the other address fields. The stage subspaces are disjoint, so no factor for the number of future locations is needed. The argument allows superpositions of stages and seeds.

Proof of Theorem A. The upper bound is Proposition 2.1. In block t, O^* and O_t differ only at the current H-address a_t and future programmed addresses. Lemma 3.4 and (13) therefore bound the expected query mass on differing locations by $\varepsilon_N + 2^{-\lambda}$ at every oracle call. If $S \le NL$ is the circuit size, Lemma 3.2 yields

$$\mathbb{E}\ \mathrm{D}_{\mathrm{tr}}(\rho_{\mathrm{real}}, \rho_{\mathrm{comp}}) \le 2S\sqrt{\varepsilon_N + 2^{-\lambda}}. \tag{14}$$

Because $r = \lambda = 100 \log N$ and $100 \log(5/4) > 32$, the expression under the square root is at most N^{-30} for sufficiently large N. Also $S \le N^2$, so (14) is at most $2N^{-13} \le N^{-10}$.

The comparison execution never uses s_N: its last snapshot still leaves the last H-transition at its base value. Its output is consequently independent of $(s_N)_1$ and guesses that bit with probability exactly $1/2$. For each fixed planting outcome, trace distance bounds the change in the event "output equals $(s_N)_1$." Average this inequality and use (8). The real execution has success at most $1/2 + N^{-10}$, and Lemma 3.3 identifies its input–oracle distribution with the required independent uniform one.

Finally, $L \ge N/(32 \cdot 301 \log N)$ for sufficiently large N, so any $c \le 1/(32 \cdot 301)$ suffices. □

Corollary 3.5 (No generic subquadratic compilation)**.** *For any fixed $\delta > 0$, no oracle-independent transformation converts every size-T RAG oracle circuit into an ordinary quantum oracle circuit of size $T^{2-\delta}\,\mathrm{polylog}\,T$ while preserving bounded-error correctness on every input and every oracle table.*

Proof. Apply the transformation to Proposition 2.1, with $T = O(N \log N)$. Its output would have size $o(N^2/\log N)$ and average success at least $2/3$, contradicting Theorem A. □

PART II

QRAM-free traveling salesman

4 The fixed-family principle

A full chain in $\mathcal{F} \subseteq 2^{[n]}$ is

$$\varnothing = A_0 \subset A_1 \subset \cdots \subset A_n = [n], \qquad |A_j| = j, \quad A_j \in \mathcal{F}.$$

Let $C(\mathcal{F})$ count full chains, and write

$$M = |\mathcal{F}|, \qquad p = p(\mathcal{F}) = \frac{C(\mathcal{F})}{n!} > 0. \tag{15}$$

A permutation is *supported* by $\mathcal{F}$ when all its prefix sets lie in $\mathcal{F}$. Thus p is exactly the fraction of supported permutations. The family is fixed independently of the edge weights.

For a directed weight matrix W, let $\Phi_{\mathcal{F}}(W)$ be the minimum tour weight among supported permutations, including the closing edge; put $\Phi_{\mathcal{F}}(W) = +\infty$ if none is feasible. We handle $n \le 1$ directly and assume $n \ge 2$ below. Missing edges have infeasibility flags. A common shift of all finite weights makes them nonnegative and changes every finite tour weight by the same known amount.

Theorem 4.1 (QRAM-free set-family principle)**.** *For a fixed family of size M and positive full-chain density p, directed TSP on explicit polynomial-bit weight matrices can be solved with bounded error using*

$$O^*(M/\sqrt{p}) \text{ ordinary gates} \qquad \text{and} \qquad O^*(M) \text{ qubits}. \tag{16}$$

The algorithm outputs an optimum tour and its value, or reports infeasibility. Its gates are exact over a fixed finite gate set. The reusable dynamic-programming circuit can be constructed from the family list in $O^*(M)$ classical time; writing out the full amplified circuit takes $O^*(M/\sqrt{p})$ time.

The proof has two ingredients: a coherent evaluation circuit whose exponential dependencies are fixed, and a dense set of input relabelings for which that circuit returns the true optimum.

4.1 A dynamic program with fixed wiring

Lemma 4.2 (Static evaluation)**.** *There is a clean reversible circuit of $O^*(M)$ gates and qubits for $\Phi_{\mathcal{F}}(W)$ and a minimizing supported tour. Its wiring depends only on $(n, \mathcal{F})$, not on W, and it can be constructed from the list of $\mathcal{F}$ in $O^*(M)$ time.*

Proof. For every starting vertex a, let $D_a(A, v)$ be the minimum weight of a path from a to v visiting exactly A, all of whose prefix sets belong to $\mathcal{F}$. Initialize $D_a(\{a\}, a) = 0$ when $\{a\} \in \mathcal{F}$. All invalid states and all states with a missing set are infeasible. For $|A| \ge 2$ and $v \ne a$, use

$$D_a(A, v) = \min_{u \in A \setminus \{v\}} \big(D_a(A \setminus \{v\}, u) + w_{uv}\big). \tag{17}$$

Then

$$\Phi_{\mathcal{F}}(W) = \min_{a \ne v} \big(D_a([n], v) + w_{va}\big). \tag{18}$$

Removing the last vertex proves (17) by induction; adding the closing edge proves (18). Infeasibility flags propagate through both operations.

There are at most n^2M states and n candidates per state. Each addition, comparison, and selection costs $\mathrm{poly}(n, b)$ Boolean gates. Crucially, every predecessor location in (17) is determined by $(\mathcal{F}, A, v, u)$ before the weights are supplied. Replace each reference by a fixed wire. To obtain a tour without addressed backtracking, carry an $O(n \log n)$-bit path description with each finite value and select it along with that value, breaking ties deterministically. The total circuit size remains $O^*(M)$.

Make the circuit reversible by computing each Boolean gate into a fresh qubit, retaining intermediate results, copying the designated outputs, and reversing the computation. This gives $O^*(M)$ gates and qubits, valid on superpositions of weight matrices by linearity. It does not clone arbitrary quantum states.

For circuit construction, enumerate the polynomially many state and transition records per set. Sort them by predecessor labels and merge against the state list to resolve every fixed connection. Labels have polynomial length and $\log M \le n$, so the work is $O^*(M)$. The same ordered list computes $C(\mathcal{F})$: count paths in the one-element-extension graph of sets. The counts have at most $O(n \log n)$ bits. □

4.2 Relabel the input, not the memory layout

For $\pi \in S_n$, set

$$W^{\pi}_{uv} = W_{\pi(u),\pi(v)}.$$

Constructing W^{π} has a polynomial-size ordinary circuit: for each fixed (u, v), test every pair (a, b) and conditionally copy the fixed input entry W_{ab} when $\pi(u) = a$ and $\pi(v) = b$. This acts on only a polynomial-size matrix. Literal multiplexers suffice.

Compute W^{π}, apply Lemma 4.2, copy the result, and uncompute. We obtain a clean circuit of $O^*(M)$ gates and qubits implementing

$$|\pi, W, z, 0\rangle \longmapsto |\pi, W, z \oplus \Phi_{\mathcal{F}}(W^{\pi}), 0\rangle. \tag{19}$$

An infeasibility flag is part of the output encoding. The DP layout is identical for every π. Its exponential workspace may become entangled with π, but it is always at known wire locations.

Lemma 4.3 (Success density)**.** *Every restricted tour is a genuine tour, so $\Phi_{\mathcal{F}}(W^{\pi}) \ge \mathrm{OPT}(W)$. If W has a tour, then*

$$\mathbb{P}_{\pi \in S_n}[\Phi_{\mathcal{F}}(W^{\pi}) = \mathrm{OPT}(W)] \ge p. \tag{20}$$

Proof. Fix an ordering σ of one optimal tour. In the relabeled graph its ordering is $\pi^{-1} \circ \sigma$, which is uniform on S_n. Exactly a p-fraction of such orderings are supported. On that event the restricted optimum equals the true optimum. □

4.3 Exact search on padded permutation ranks

There is no need to prepare an exact uniform superposition on $n!$ labels. Put

$$\ell = \lceil \log(n!) \rceil$$

and apply Hadamards to ℓ zero qubits. Interpret the resulting binary string y as an integer. When $y < n!$, let π_y be the permutation of rank y; otherwise declare the label invalid. For a finite threshold B, mark

$$f_B(y) = \mathbf{1}\{y < n! \text{ and } \Phi_{\mathcal{F}}(W^{\pi_y}) \le B\}. \tag{21}$$

Rank-to-permutation conversion has a polynomial-size Boolean circuit: successively divide by factorials and choose the indicated element from the sorted list of remaining vertices. Each variable-index operation concerns only n labels and can be replaced by an ordinary multiplexer. For invalid ranks, use any fixed dummy permutation and force the mark to zero.

The marking circuit is therefore exact, clean, and of size $O^*(M)$. By Lemma 4.3, its accepting fraction under uniform ℓ-bit strings is

$$\begin{cases} 0, & \mathrm{OPT}(W) > B, \\ \geq \dfrac{n!}{2^\ell} p \geq p/2, & \mathrm{OPT}(W) \leq B. \end{cases} \tag{22}$$

The elementary search lemma proved in Appendix A finds a marked label, when this fraction is at least $p/2$, using $O(p^{-1/2}\log(1/\delta))$ evaluations for failure probability δ. Invalid ranks only cost a constant factor. There are no arbitrary rotations or precision parameters.

Proof of Theorem 4.1. After shifting finite weights, let $B_{\max}$ be an integer upper bound on every finite tour value; it has $O(b + \log n)$ bits. First run the search at $B_{\max}$. If it finds no marked rank, report no tour. On an infeasible instance this is certain; on a feasible instance failure has the chosen small probability.

Otherwise binary-search the integer interval $[0, B_{\max}]$, using (21)–(22) for every threshold. There are $K = O(b + \log n + 1)$ tests. Give each test error at most $1/(20K)$, including the initial feasibility test. The guarantee holds for each threshold even conditional on earlier outcomes, so a union bound gives the exact optimum value with probability at least $19/20$.

At that threshold, one further search with failure at most $1/20$ finds a valid rank whose restricted optimum is the true optimum. Measure the rank, compute the witness-producing DP, and relabel the resulting tour back by π_y. Undo the initial common weight shift. The total success probability is at least $9/10$, hence at least $2/3$.

Each search evaluation costs $O^*(M)$ gates and returns its workspace to zero. Polynomially many tests, repetitions, and control registers give (16); there are not $p^{-1/2}$ copies of the DP memory. All arithmetic, phase marking, Hadamards, and reflections are exact over a fixed finite gate set, as detailed in Appendix A.

Lemma 4.2 constructs the reusable template in $O^*(M)$ time. The full amplified gate list has length $O^*(M/\sqrt{p})$ and can be generated in that much time by repeated emission of the template. A streaming generator need retain only the template and polynomial-size loop counters. Gate locations remain fixed independently of W and of measurement outcomes; fixed-location controls implement the latter. □

5 The numerical bound and uniformity

The combinatorial input can be stated entirely in terms of one finite family.

Theorem 5.1 (External finite-family input)**.** *There exist a positive integer d and a family $\mathcal{A} \subseteq 2^{[d]}$ with $C(\mathcal{A}) > 0$ such that*

$$|\mathcal{A}|^{1/d} < 1.430, \qquad \left(\frac{|\mathcal{A}|}{\sqrt{p(\mathcal{A})}}\right)^{1/d} < 1.784. \tag{23}$$

This follows from Andoni–Dallant–Kozma–Yu [12, Theorems 1.1 and 4.4]. Their entropy characterization and numerical witness give, for every $\epsilon > 0$, a finite family with

$$\frac{\log|\mathcal{A}|}{d} \le 0.515617 + \epsilon, \qquad \frac{-\log p(\mathcal{A})}{d} \le 0.638624 + \epsilon.$$

Taking $\epsilon = 10^{-5}$ gives

$$2^{0.515627} < 1.429616 < 1.430, \qquad 2^{0.834944} < 1.783788 < 1.784,$$

which proves (23). The positive margin avoids assuming attainment of an extremal supremum or suppressing an unspecified $2^{o(n)}$ factor. The external numerical witness, rather than a new extremal construction, supplies these constants.

For families on disjoint ground sets, let their *union product* consist of all unions of one constituent set from each factor. Then

$$|\mathcal{F}_1 \otimes \mathcal{F}_2| = |\mathcal{F}_1||\mathcal{F}_2|, \qquad p(\mathcal{F}_1 \otimes \mathcal{F}_2) = p(\mathcal{F}_1)p(\mathcal{F}_2). \tag{24}$$

The first identity is immediate. For the second, a supported ordering is an interleaving of one supported ordering from each factor. If the ground-set sizes are n_1, n_2, the chain count is

$$\binom{n_1 + n_2}{n_1} C(\mathcal{F}_1)C(\mathcal{F}_2).$$

Divide by $(n_1 + n_2)!$. These product identities also appear in the set-system framework [11, Lemma 3.7].

Proof of Theorem B. Fix one finite witness $\mathcal{A}$ from Theorem 5.1. Write $n = kd + q$ with $0 \le q < d$, and take

$$\mathcal{F}_n = \mathcal{A}^{\otimes k} \otimes 2^{[q]},$$

placing factors on disjoint blocks. The remainder factor has size 2^q and density one. By (24),

$$|\mathcal{F}_n| = |\mathcal{A}|^k 2^q = O(1.430^n), \qquad \frac{|\mathcal{F}_n|}{\sqrt{p(\mathcal{F}_n)}} = \left(\frac{|\mathcal{A}|}{\sqrt{p(\mathcal{A})}}\right)^k 2^q = O(1.784^n).$$

The constants depend on the fixed d, not on n. Apply Theorem 4.1.

This is uniform. The finite list of $\mathcal{A}$ may be part of one algorithm's fixed description, and its product family can be enumerated in polynomial time per output set. Alternatively, find a witness once by enumerating finite families and testing the strict rational inequalities

$$|\mathcal{A}| < (1.430)^d, \qquad \frac{|\mathcal{A}|^2 d!}{C(\mathcal{A})} < (1.784)^{2d}. \tag{25}$$

Their truth is decidable after clearing denominators; Theorem 5.1 guarantees termination. This one-time search costs a constant independent of the TSP input size. The resulting density $p(\mathcal{F}_n) = p(\mathcal{A})^k$ is an exactly computable rational with $O(n)$-bit numerator and denominator. It determines the search cap, while the family list determines all fixed DP wires.

Constructing the reusable circuit costs $O^*(|\mathcal{F}_n|)$ time; generating the full gate sequence costs $O^*(|\mathcal{F}_n|/\sqrt{p(\mathcal{F}_n)})$. Thus uniform generation respects the stated running-time bound. The construction can hide an enormous constant, but neither an input-dependent advice table nor an uncharged QRAM. □

6 Relation to prior work and interpretation

Oracle separation. Chen–Wang [1] prove a classical RAM versus multitape-Turing-machine separation with short random oracles; their formal Theorem 4.3 is distributional over the input and oracle. Here the competing machine is an arbitrary fixed-wiring quantum circuit, and the hard function has a separate address sampler. The accessible part of a tape is replaced by the small reduced state on the wires touched by a circuit block. Slicing and direct-product arguments have precedents in quantum time–space tradeoffs [5]. The point here is not a total-space bound, but a bound on the memory a short fixed block can touch.

Quantum hash-chain sequentiality is studied using compressed-oracle techniques [6]; it does not by itself charge the cost of reading explicit input data. Our proof instead uses fixed snapshots and the random-access-code theorem. Likewise, a lower bound for implementing one lookup does not automatically give a lower bound for the final result of a computation: a competing circuit may reorganize the work. This gate-count versus implementation distinction is discussed in the QRAM survey [8].

TSP. Ambainis et al. obtain the approximately 1.728^n QRAM bound using exponential dynamic programming [9]; subsequent work studies its time–space tradeoffs [10]. The restricted-family DP comes from the extremal-set-system approach [11, 12]. Dallant's quantum version has the same $M/\sqrt{p}$ scale and is stated with quantum read-write QRAM [13, Sections 2.4 and 3]. Part II keeps that scale while fixing one DP layout, coherently relabeling its input, and searching padded ranks rather than a stored list of family members. The contribution under consideration is QRAM removal, not a new chain-density bound.

Why the results do not give a TSP separation. The oracle upper bound uses only classical addressed reads. In models with a free coherent input-query gate, such as the input-access convention of [7, Section 2], the hard function is easy even without working-memory RAGs. Also, Part I begins with N independent input bits, whereas the entire TSP table is determined by $O(n^2 b)$ input bits. Its exponential number of entries cannot be substituted for N in the random-access-code argument. Part II gives a concrete alternative to those table lookups: coherent recomputation on exponential fixed-wire workspace. Neither theorem asserts a depth separation or a quadratic QRAM advantage for TSP.

A Exact search with a known density lower bound

This is the standard amplitude-amplification argument [4], specialized to uniform binary labels so that no approximate rotations are needed.

Lemma A.1 (Dense search)**.** *Let $f : \{0,1\}^\ell \to \{0,1\}$ have a clean evaluator over NOT, CNOT, and Toffoli gates. If its accepting fraction is either zero or at least a known $p_0 \in (0,1]$, a bounded-size procedure finds a marked label with probability at least $1-\delta$ in the latter case and never returns an unmarked label. It uses*

$$O(p_0^{-1/2}\log(1/\delta))$$

evaluator calls and inverses, with polynomial overhead per call in ℓ and $\log(1/p_0)$, and polynomial additional workspace in ℓ, $\log(1/p_0)$, and $\log(1/\delta)$. A fixed finite bounded-arity gate set suffices exactly.

Proof. Prepare $|+\rangle^{\otimes\ell}$. Write the accepting fraction as $a = \sin^2\theta$. A Grover iteration is a marked-state phase flip followed by reflection about this initial state. After j iterations the success probability is $\sin^2((2j+1)\theta)$.

Let R be the smallest power of two at least $2/\sqrt{p_0}$, so $R < 4/\sqrt{p_0}$. In one trial, with probability $1/2$ use the initial state directly; otherwise choose j uniformly from $\{0,\dots,R-1\}$ and apply j iterations. Measure and verify the label with f.

If $a \ge 3/4$, the direct branch alone succeeds with probability at least $3/8$. If $p_0 \le a \le 3/4$, then $\sin(2\theta) = 2\sqrt{a(1-a)} \ge \sqrt{p_0}$, and

$$\frac{1}{R}\sum_{j=0}^{R-1}\sin^2((2j+1)\theta) = \frac{1}{2} - \frac{\sin(4R\theta)}{4R\sin(2\theta)} \ge \frac{3}{8}.$$

Thus a trial succeeds with probability at least $3/16$. Repeating $O(\log(1/\delta))$ times gives the claim; verification rejects every label when $a = 0$.

For a fixed schedule, retain the random branch and j and unroll all R iterations. In iteration i, compute an active flag for "amplified branch and $i \le j$." Compute f unconditionally, apply its phase only when active, and uncompute. Implement the reflection as $H^{\otimes\ell}(2|0^\ell\rangle\langle 0^\ell| - I)H^{\otimes\ell}$, controlling only the central phase on the active flag. That phase is -1 exactly when the label is nonzero and the iteration is active; compute these conditions reversibly into a flag. When inactive, the whole iteration is the identity.

All random bits and initial superpositions use Hadamards; the predicates and central phases use X, Z, CNOT, and Toffoli. Workspace is uncomputed and reused. The comparisons and reflections have polynomial cost in $\ell + \log R$. Repeating trials needs only label and control records, not separate copies of the evaluator workspace; measurements can be deferred on those specified registers. □

In Part II, $p_0 = p/2$ and $p \ge 1/n!$, so ℓ and $\log(1/p_0)$ are $O(n\log n)$. All additional work is absorbed by O^*. Thresholds can likewise be stored and updated on fixed registers. Their values affect local controls, not quantum memory addresses.

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